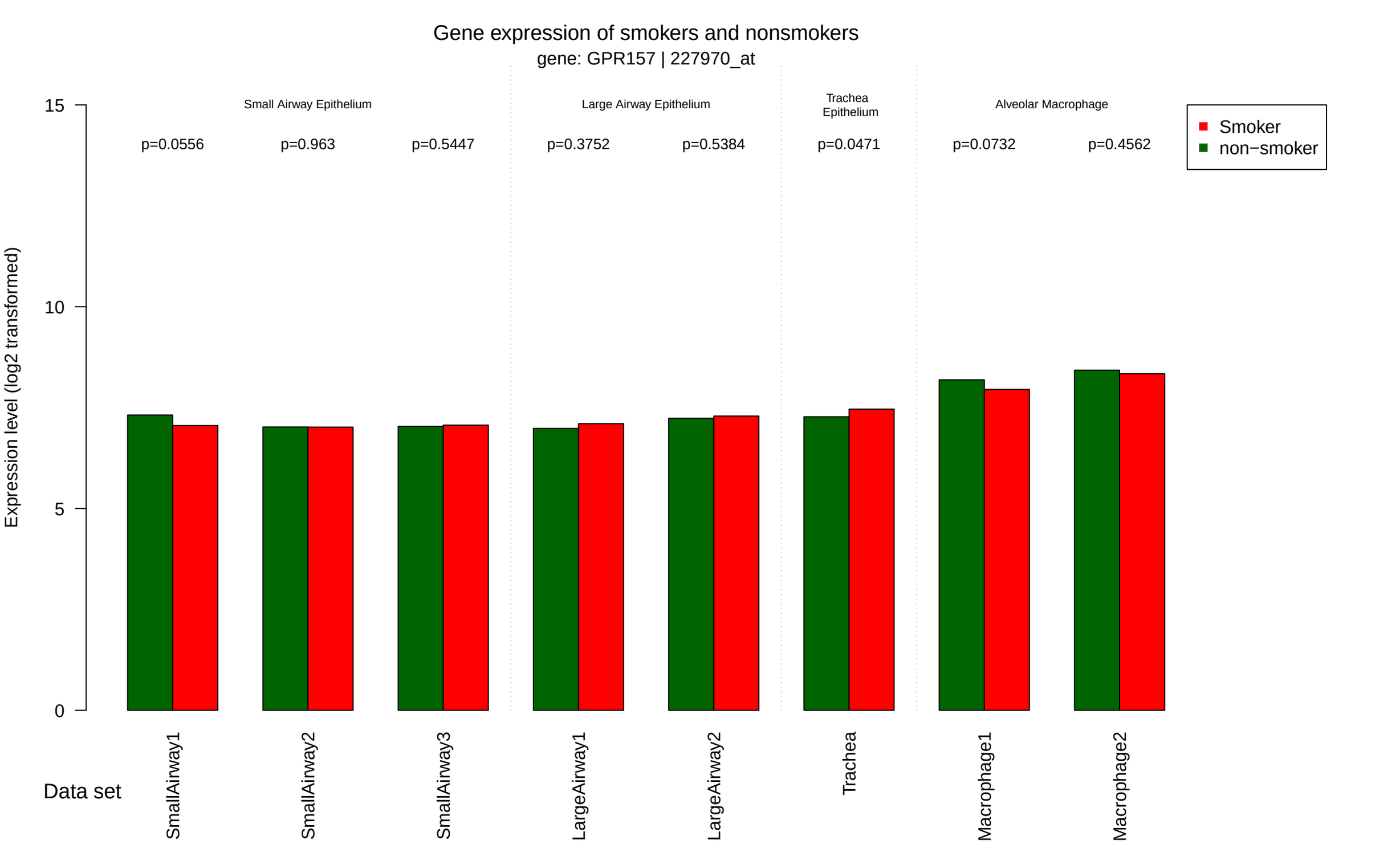
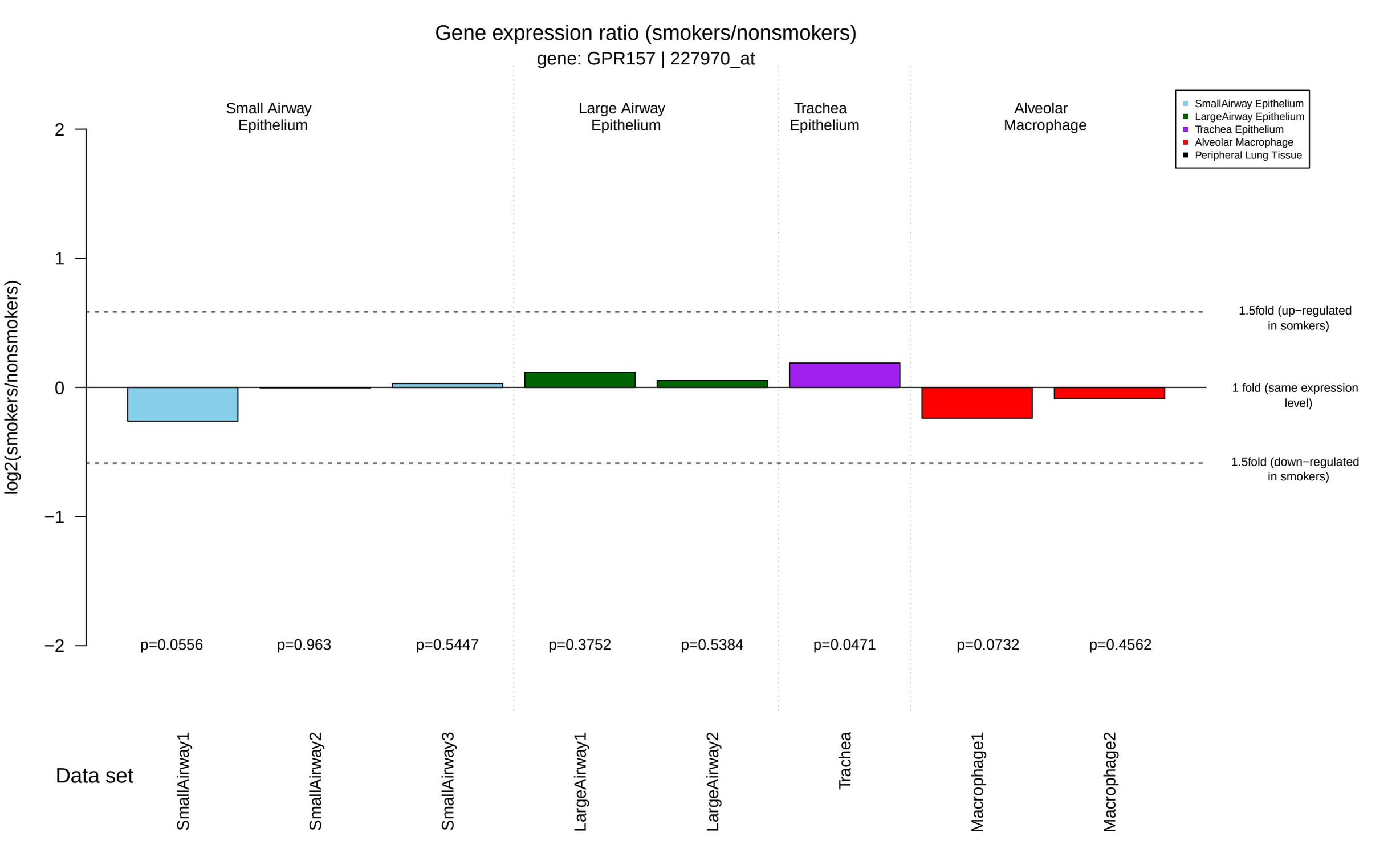
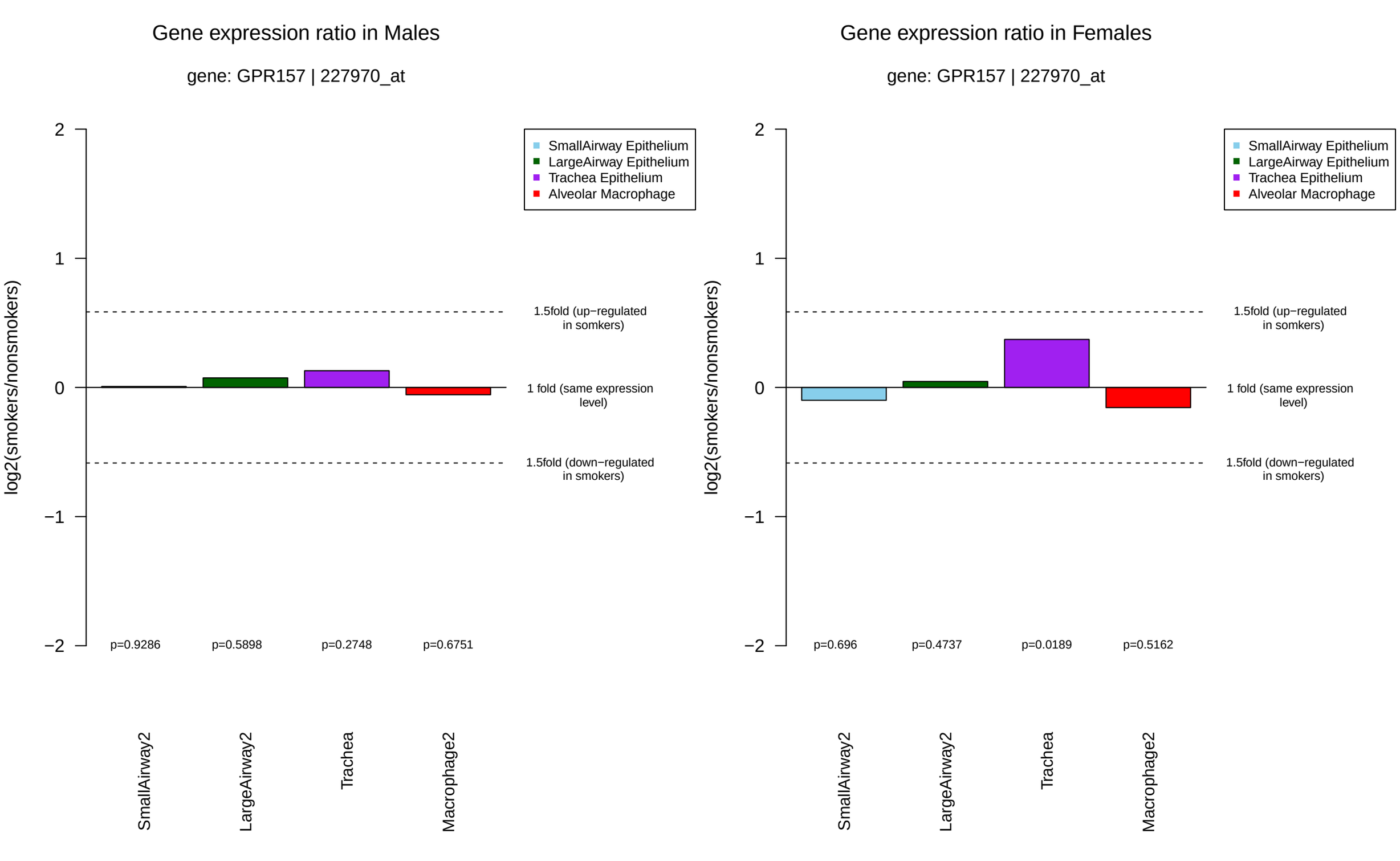
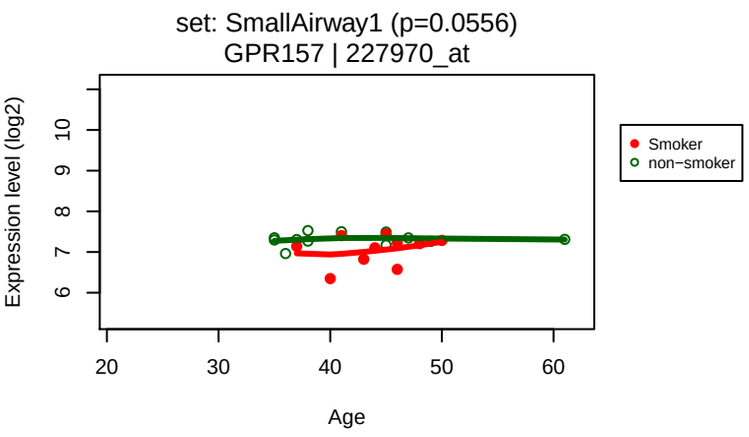
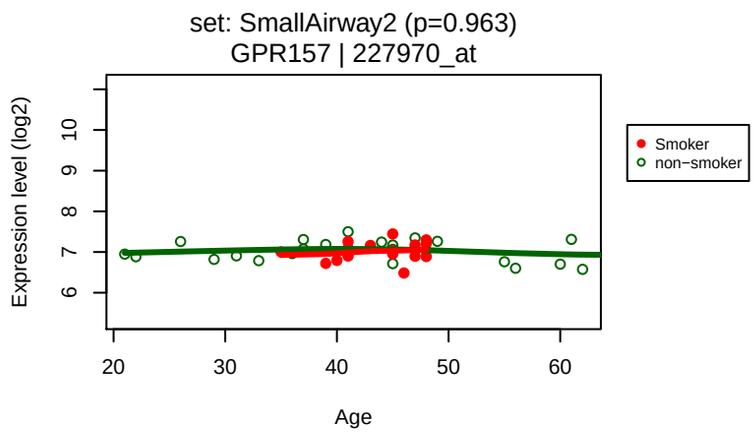
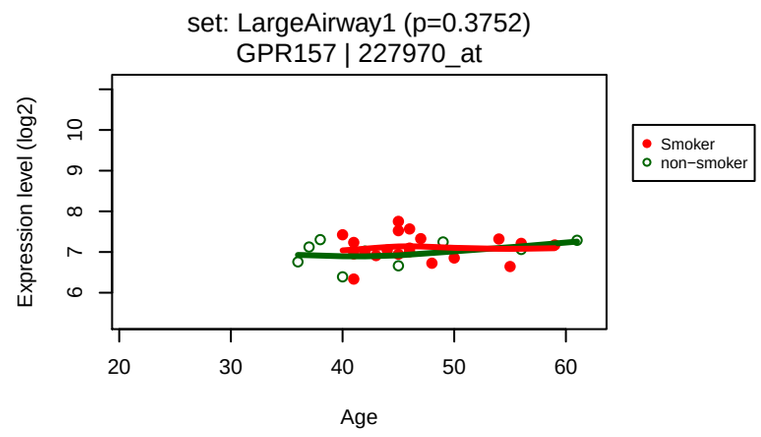

Gene expression of smokers and nonsmokers
gene: GPR157 | 227970_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0556
p=0.963
p=0.5447
p=0.3752
p=0.5384
p=0.0471
p=0.0732
p=0.4562
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

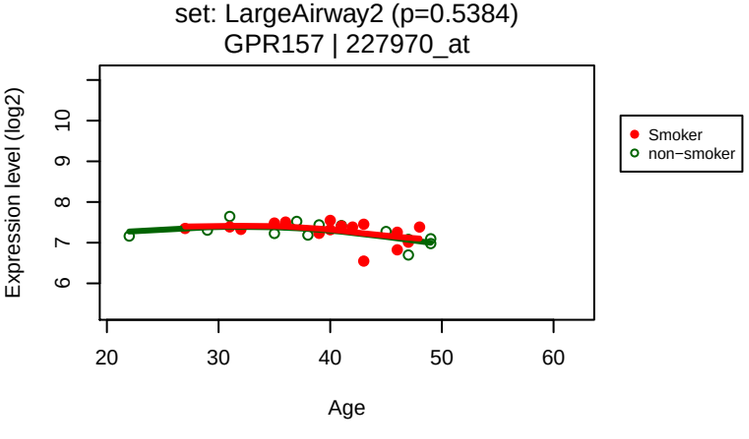

Gene expression ratio (smokers/nonsmokers)
gene: GPR157 | 227970_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0556
p=0.963
p=0.5447
p=0.3752
p=0.5384
p=0.0471
p=0.0732
p=0.4562
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

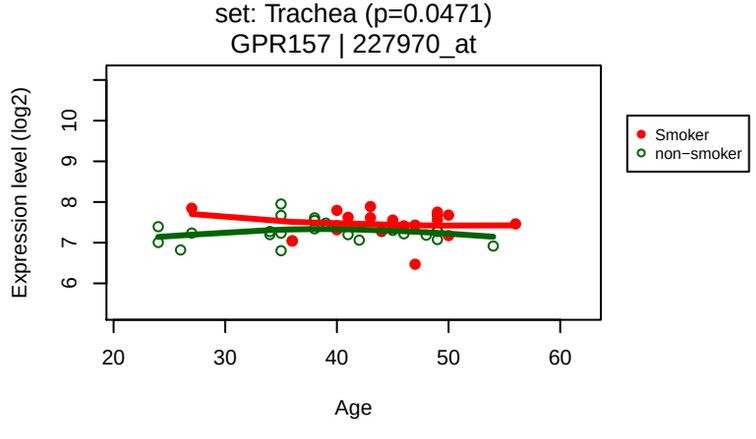

Gene expression ratio in Males
gene: GPR157 | 227970_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9286
p=0.5898
p=0.2748
p=0.6751
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPR157 | 227970_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.696
p=0.4737
p=0.0189
p=0.5162
SmallAirway2
LargeAirway2
Trachea
Macrophage2

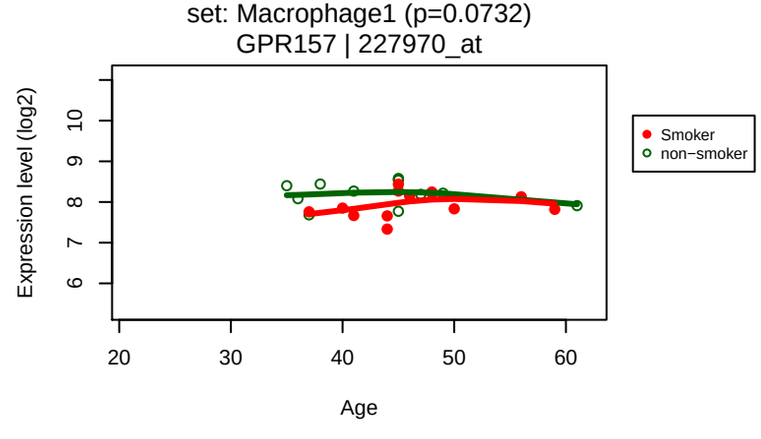

set: SmallAirway1 (p=0.0556)
GPR157 | 227970_at
Expression level (log2)
Age
Smoker
non−smoker

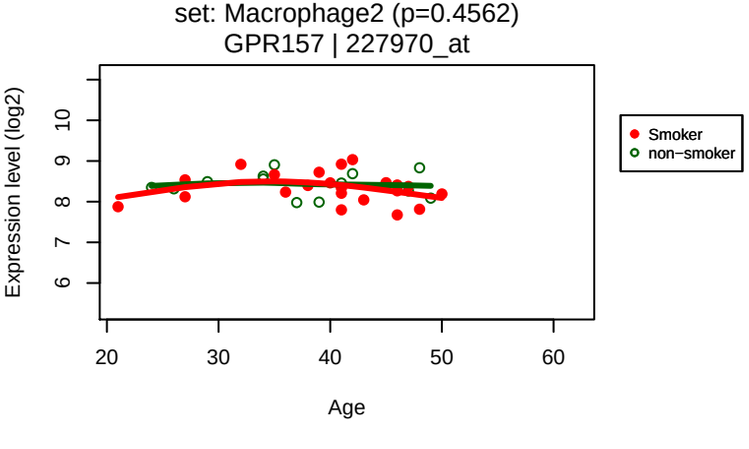

set: SmallAirway2 (p=0.963)
GPR157 | 227970_at
Expression level (log2)
Age
Smoker
non−smoker

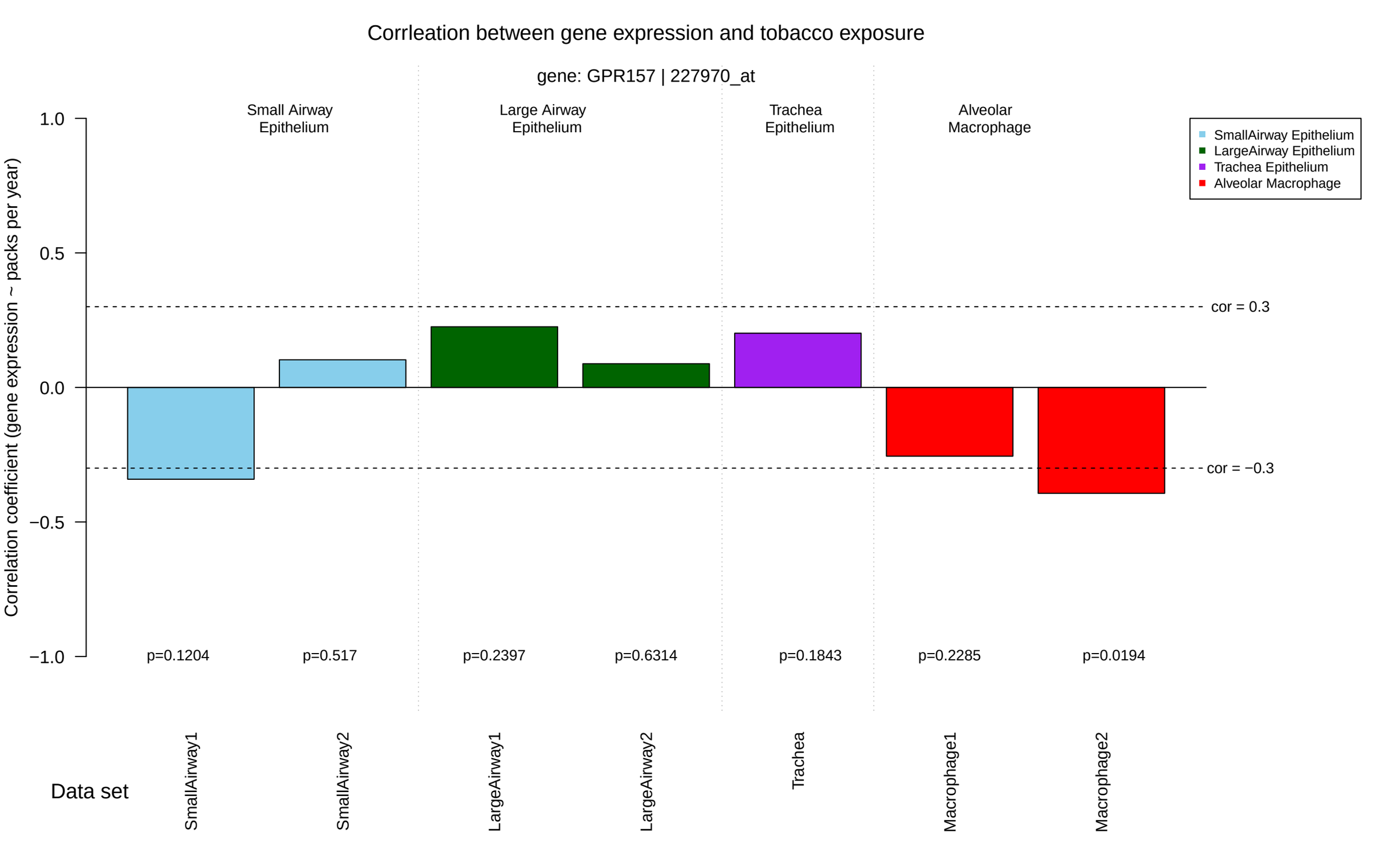

set: LargeAirway1 (p=0.3752)
GPR157 | 227970_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5384)
GPR157 | 227970_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0471)
GPR157 | 227970_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0732)
GPR157 | 227970_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4562)
GPR157 | 227970_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPR157 | 227970_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1204
p=0.517
p=0.2397
p=0.6314
p=0.1843
p=0.2285
p=0.0194
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2